

Homework 4

Meenu Ravi

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1 Performing solution verification studies

I have updated my code from Homework 3 such that the 2 SRQs I am observing are temperature and fuel amount over the space. My domain is the grid of 1 x 1 where there is high temperature in the middle at (0.5,0.5) and near 0 else where. Fuel is initialized to 1 across the entire domain to represent a forest full of vegetation. The fire spreads outward over time as it burn through vegetation. Specifically, my SRQs are maximum temperature across the domain at the final time and total remaining fuel amount at final time. The code for this is in <https://github.com/meenuravi18/Verification-and-Validation-in-Scientific-Computing/tree/main/homework4>

2 Quantifying the effects of discretization error

I use atleast four systematically refined grids of refinement factor $r = 2$ to estimate the discretization error via Richardson Extrapolation. The numbers here have been round for space constraints.

Observed Order of Accuracy (Richardson Extrapolation) for temperature:

$$p_1 = \frac{\ln\left(\frac{f_{t1}-f_{t2}}{f_{t2}-f_{t3}}\right)}{\ln(r)}$$

$$p_2 = \frac{\ln\left(\frac{f_{t2}-f_{t3}}{f_{t3}-f_{t4}}\right)}{\ln(r)}$$

solving for both:

$$p_1 = 2.10726$$

$$p_2 = 2.05313$$

Observed Order of Accuracy (Richardson Extrapolation) for fuel:

$$p_3 = \frac{\ln\left(\frac{f_{f1}-f_{f2}}{f_{f2}-f_{f3}}\right)}{\ln(r)}$$

$$p_4 = \frac{\ln\left(\frac{f_{f2}-f_{f3}}{f_{f3}-f_{f4}}\right)}{\ln(r)}$$

solving for both:

$$p_3 = 2.1048$$

$$p_4 = 2.0525$$

where $f_{t1}, f_{t2}, f_{t3}, f_{t4}$ are SRQ values on systematically refined grids from coarse to fine for temperature and $f_{f1}, f_{f2}, f_{f3}, f_{f4}$ are SRQ values on systematically refined grids from coarse to fine for fuel.

2.1 Fine Grid

Factor of Safety

$$\text{Temperature : } F_{ts} = \begin{cases} 1.25 & \text{if } \frac{|(p_2)-2|}{2} \leq 0.1 \\ 3.0 & \text{if } \frac{|(p_2)-2|}{2} > 0.1 \end{cases}$$

$$\text{Fuel : } F_{fs} = \begin{cases} 1.25 & \text{if } \frac{|(p_4)-2|}{2} \leq 0.1 \\ 3.0 & \text{if } \frac{|(p_4)-2|}{2} > 0.1 \end{cases}$$

The factor of safety for both temperature and fuel were 1.25.

2.1.1 Uncertainty due to Discretization error

Roache's Grid Convergence Index (GCI)

$$GCI_{temp} = \frac{F_{ts}}{r^{p_2} - 1} |f_{t4} - f_{t3}|$$

$$GCI_{fuel} = \frac{F_{fs}}{r^{p_4} - 1} |f_{f4} - f_{f3}|$$

where F_s is the Factor of Safety

Uncertainty Due to Discretization Error- Fine Grid GCI Values

$$GCI_{temp} = 0.00027$$

$$GCI_{fuel} = 0.00018$$

$$\tilde{f} \in [f_h - GCI_{temp}, f_h + GCI_{temp}] \quad (1)$$

For temperature:

$$\tilde{f} \in [f_{th4} - 0.00027, f_{th4} + 0.00027] \quad (2)$$

where $f_{th4} = 1.89251$

For fuel:

$$\tilde{f} \in [f_{fh4} - 0.000186, f_{fh4} + 0.000186] \quad (3)$$

where $f_{fh4} = 1.9998$

For temperature:

$$\tilde{f} \in [1.89251 - 0.00027, 1.89251 + 0.00027] \quad (4)$$

For fuel:

$$\tilde{f} \in [1.9998 - 0.000186, 1.9998 + 0.000186] \quad (5)$$

2.1.2 Discretization Error Estimates

Discretization Error Estimate for temperature:

$$\bar{f} = \frac{f_{t3} - f_{t4}}{2^{p_2} - 1} \quad (6)$$

Discretization Error Estimate for fuel:

$$\bar{f} = \frac{f_{f3} - f_{f4}}{2^{p_4} - 1} \quad (7)$$

Discretization Error- Fine Grid

$$\begin{aligned} DE_{temp} &= -0.00022 \\ DE_{fuel} &= -0.00015 \end{aligned}$$

2.2 Coarse Grid

Factor of Safety

$$\text{Temperature : } F_{ts} = \begin{cases} 1.25 & \text{if } \frac{|(p_1)-2|}{2} \leq 0.1 \\ 3.0 & \text{if } \frac{|(p_1)-2|}{2} > 0.1 \end{cases}$$

$$\text{Fuel : } F_{fs} = \begin{cases} 1.25 & \text{if } \frac{|(p_3)-2|}{2} \leq 0.1 \\ 3.0 & \text{if } \frac{|(p_3)-2|}{2} > 0.1 \end{cases}$$

The factor of safety for both temperature and fuel were 1.25.

2.2.1 Uncertainty due to Discretization error

Roache's Grid Convergence Index (GCI)

$$GCI_{temp} = \frac{F_{ts}}{r^{p_1} - 1} |f_{t2} - f_{t1}|$$

$$GCI_{fuel} = \frac{F_{fs}}{r^{p_3} - 1} |f_{f2} - f_{f1}|$$

where F_s is the Factor of Safety

Uncertainty due to Discretization- Coarse Grid GCI Values

$$\begin{aligned} GCI_{temp} &= 0.00471 \\ GCI_{fuel} &= 0.00316 \end{aligned}$$

$$\tilde{f} \in [f_h - GCI_{temp}, f_h + GCI_{temp}] \quad (8)$$

For temperature:

$$\tilde{f} \in [f_{th2} - 0.00471, f_{th2} + 0.00471] \quad (9)$$

where $f_{th2} = 1.8889$

For fuel:

$$\tilde{f} \in [f_{fh2} - 0.00316, f_{fh2} + 0.00316] \quad (10)$$

where $f_{fh2} = 1.99743$

Substituted, it becomes:

For temperature:

$$\tilde{f} \in [1.8889 - 0.00471, 1.8889 + 0.00471] \quad (11)$$

For fuel:

$$\tilde{f} \in [1.99743 - 0.00316, 1.99743 + 0.00316] \quad (12)$$

2.2.2 Discretization Error Estimates

Discretization Error Estimate for temperature:

$$\bar{f} = f_{t3} + \frac{f_{t3} - f_{t2}}{2^{p_1} - 1}$$
$$DE_{temp} = f_{t2} - \bar{f}$$

Discretization Error Estimate for fuel:

$$\bar{f} = f_{f3} + \frac{f_{f3} - f_{f2}}{2^{p_3} - 1}$$
$$DE_{fuel} = f_{f2} - \bar{f} \tag{13}$$

Discretization Error- Coarse Grid

$$DE_{temp} = -0.00377$$
$$DE_{fuel} = -0.00253$$

3 Quantifying the effects of iterative error for SRQs

For this work, iterative error is not applicable because the solver is RK4 with fixed time steps rather than an iterative convergence.

4 Quantifying the effects of round-off error for SRQs

For estimating the round-off error, I found from homework 3 that using float64 and increasing to np.longdouble, both defaulted to float64 (in python MacOS), resulting in a round-off estimate of 0. Therefore, I used float32 (single-precision) and float64 (double-precision) as the low and high precisions respectively. The simulation was run at both precisions with the same mesh and time step, and the round-off error was estimated as the difference between the two solutions for fuel and temperature respectively. The errors were:

Mesh (16) – most coarse

- Temperature: $2.4721e - 08$
- Fuel: $7.2199e - 08$

Mesh (32)

- Temperature: $3.64463e - 08$
- Fuel: $7.36024e - 08$

Mesh (64)

- Temperature: $8.41772e - 08$
- Fuel: $1.78461e - 08$

Mesh (128) – most fine

- Temperature: $1.01701e - 07$
- Fuel: $2.58478e - 08$

Since I don't have iterative error, I am not able to check if this iteratively converges to machine zero and check if round-off error dominates iterative error. However, it was said that both should be estimated to be atleast 100 times smaller than the discretization error on the finest mesh and time step.

5 Total Numerical uncertainty

Fine Grid:

$$\begin{aligned}U_{num} &= U_{DE} + U_{IE} + U_{RO} \\U_{temp} &= 0.00027 + 1.01701e - 07 \\U_{fuel} &= 0.00018 + 2.58478e - 08\end{aligned}$$

Coarse Grid:

$$\begin{aligned}U_{num} &= U_{DE} + U_{IE} + U_{RO} \\U_{temp} &= 0.00471 + 2.4721e - 08 \\U_{fuel} &= 0.00316 + 7.2199e - 08\end{aligned}$$